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DATA SET DS1509

9. Refer to Exercise 4 (Section 15.2). The data in this table are from a paired-difference experiment with $n = 7$ pairs of observations.

	Pair						
Population	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9

a. Use Wilcoxon's signed-rank test to determine whether there is a significant difference between the two populations.

b. Compare the results of part a with the result you got in Exercise 4 (Section 15.2). Are they the same? Explain.

a) H_0: There is no significant difference between two populations.
H_a: There is significant difference between two populations.

	Pair						
Population	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9
d	0.1	0.7	0.3	-0.1	0.5	0.2	0.5
$\lvert d \rvert$	0.1	0.7	0.3	0.1	0.5	0.2	0.5
Rank	1.5	7	4	1.5	5.5	3	5.5
Sign	+	+	+	-	+	+	+

$T^+ = 1.5 + 7 + 4 + 5.5 + 3 + 5.5 = 26.5$

$T^- = 1.5$

$T = \text{Min}(T^+, T^-)$

$= 1.5$

$n = 7$

RR: $T \le 3$

$\because 1.5 < 3$

$\therefore$ Reject H_0

b) Wilcoxon signed-rank test finds a significant difference, but the earlier sign test did not. The results are different because Wilcoxon uses both signs and the sizes of the differences, while the sign test uses only the signs.

11. Machine Breakdowns The number of machine breakdowns per month was recorded for 9 months on two identical machines, A and B, used to make wire rope:

DATA SET DS1511

Month	A	B	$d = A - B$		Rank
1	3	7	-4	$-$	6
2	14	12	2	$+$	3.5
3	7	9	-2	$-$	3.5
4	10	15	-5	$-$	7
5	9	12	-3	$-$	5
6	6	6	0	$+$	$-$
7	13	12	1	$+$	1.5
8	6	5	1	$+$	1.5
9	7	13	-6	$-$	8

a. Do the data provide sufficient evidence to indicate a difference in the monthly breakdown rates for the two machines? Test by using a value of α near .05.

b. Can you think of a reason the breakdown rates for the two machines might vary from month to month?

a) H_0: There is no difference in monthly breakdown rates.

H_a: There is a difference in monthly breakdown rates

$n = 8$

$T^+ = 3.5 + 1.5 + 1.5 = 6.5$

$T^- = 6 + 3.5 + 7 + 5 + 8 = 29.5$

$T = \min(T+, T-)$

$= 6.5$

RR: $T \leq 3$

$T = 6.5$

$\because 6.5 > 3$

$\therefore$ Non-reject H_0

b) The breakdown rates might vary from month to month because the machines may not be used under exactly the same conditions every month.

DATA SET DS1514 **15. Images and Word Recall** A psychology class performed an experiment to determine whether a recall score in which instructions to form images of 25 words were given differs from an initial recall score for which no imagery instructions were given. Twenty students participated in the experiment with the results listed in the table.

Student	With Imagery	Without Imagery	Student	With Imagery	Without Imagery
1	20	5 +	11	17	8 +
2	24	9 +	12	20	16 +
3	20	5 +	13	20	10 +
4	18	9 +	14	16	12 +
5	22	6 +	15	24	7 +
6	19	11 +	16	22	9 +
7	20	8 +	17	25	21 +
8	19	11 +	18	21	14 +
9	17	7 +	19	19	12 +
10	21	9 +	20	23	13 +

a. What three testing procedures can be used to test for differences in the distribution of recall scores with and without imagery? What assumptions are required for the parametric procedure? Do these data satisfy these assumptions?

b. Use both the sign test and the Wilcoxon signed-rank test to test for differences in the distributions of recall scores under these two conditions.

c. Compare the results of the tests in part b. Are the conclusions the same? If not, why not?

a) Sign test, Wilcoxon signed-rank test and paired t-test. The differences between the paired observations must be normally distributed
Likely yes. The differences show no extreme outliers, but a formal normality test is required to confirm.

b) Sign test

$H_0: p = 0.5$

$H_a: p \neq 0.5$

$n = 20$

$X \sim (20, 0.5)$

RR : $X \leq 5$ or $X \geq 15$

$X = 20$

$\because 20 \geq 15$

$\therefore$ Reject H_0

Wilcoxon signed-rank test

$n = 20$

$T^- = 0$

$T_{20} = 52$

$\because T^- = 0 < T_{20} = 52$

$\therefore$ Reject H_0

c) Yes, the results are same, there is sufficient evidence to conclude that recall scores differ between the two conditions.

6. Legos® The time required for kindergarten children to assemble a specific Lego creation was measured for children who had been instructed for four different lengths of time. Five children were randomly assigned to each instructional group. The length of time (in minutes) to assemble the Lego creation was recorded for each child in the experiment.

DATA SET DS1518

Training Period (hours)			
.5	1.0	1.5	2.0
8	9	4	4
14	7	6	7
9	5	7	5
12	8	8	5
10	9	6	4

Use the Kruskal–Wallis H-Test to determine whether there is a difference in the distribution of times for the four different lengths of instructional time. Use $\alpha = .01$.

Ho : The distributions of assembly times are the same across all 4 training periods

Ha : The distributions of assembly times are not the same.

Training Period (hours)			
.5	1.0	1.5	2.0
8 13	9 16	4 2	4 2
14 20	7 10	6 7.5	7 10
9 10	5 5	7 10	5 15
12 19	8 13	8 13	5 15
10 18	9 16	6 7.5	4 2

$T_1 = 86$ $T_2 = 60$ $T_3 = 40$ $T_4 = 24$

$$H = \frac{12}{N(N+1)}\left(\frac{T_i^2}{n_i}\right) - 3(N+1)$$

$$= \frac{12}{20(21)}\left(\frac{86^2}{5} + \frac{60^2}{5} + \frac{40^2}{5} + \frac{24^2}{5}\right) - 3(21)$$

$$= 12.27$$

$$X^2_{3,0.01} = 11.345$$

$\because H = 12.27 < X^2_{3,0.01} = 11.345$

$\therefore$ Reject Ho

DATA SET DS1520

8. Heart Rate and Exercise In Exercise 18 (Section 11.3), we presented data on the heart rates for samples of 10 men randomly selected from each of four age groups. Each man walked a treadmill at a fixed grade for a period of 12 minutes, and the increase in heart rate (the difference before and after exercise) was recorded (in beats per minute). The data are shown in the table.

	10–19	20–39	40–59	60–69
	29	24	37	28
	33	27	25	29
	26	33	22	34
	27	31	33	36
	39	21	28	21
	35	28	26	20
	33	24	30	25
	29	34	34	24
	36	21	27	33
	22	32	33	32
Total	309	275	295	282

a. Do the data present sufficient evidence to indicate differences in location for at least two of the four age groups? Test using the Kruskal–Wallis H-test with $\alpha = .01$.

b. Find the approximate p-value for the test in part a.

c. Since the F-test in Chapter 11 and the H-test are both tests to detect differences in location of the four heart-rate populations, how do the test results compare? If the ANOVA F-test produced the test statistic $F = 0.87$ with p-value $= 0.468$, compare this p-value with the p-value in part b and explain the implications of the comparison.

	10–19	20–39	40–59	60–69
	29 21	24 8	37 39	28 18
	33 29.5	27 15	25 10.5	29 21
	26 12.5	33 29.5	22 5.5	34 34
	27 15	31 24	33 29.5	36 37.5
	39 40	21 3	28 18	21 3
	35 36	28 18	26 12.5	20 1
	33 29.5	24 8	30 23	25 10.5
	29 21	34 34	34 34	24 8
	36 37.5	21 3	27 15	33 29.5
	22 5.5	32 25.5	33 29.5	32 25.5
Total	309	275	295	282
	$T_1 = 247.5$	$T_2 = 168$	$T_3 = 216.5$	$T_4 = 188$
	$n_1 = 10$	$n_2 = 10$	$n_3 = 10$	$n_4 = 10$

a) H_0: The distributions of heart-rate increases are the same for all four age groups

H_a. At least one age group has a different distribution of heart rate increases.

$$H = \frac{12}{N(N+1)} \sum \frac{T_i}{n_i} - 3(n+1)$$

$= 2.63$

$\because H < \chi^2_{3, 0.01} = 11.345$

$\therefore$ Non-reject H_0

b) $H = 2.63$ $df = 3$

$0.1 < p < 0.9$

$\because p > 0.01$

$\therefore$ Non-reject H_0

c) $F = 0.87$. p-value $= 0.468$

$\because p = 0.468 > 0.01$

$\therefore$ Non-reject H_0